

Order Flow Imbalance and Machine Learning for Short-Term Price Prediction and Market Making

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Abstract

This note reports the results of an end-to-end microstructure research pipeline built on public Binance limit order book (LOB) and trade data for SOL/USDT. We reconstruct the LOB across three trading sessions bracketing the NYSE open, show that calendar-time sampling masks latent return structure, and adopt a volume clock to mitigate this heteroscedasticity. Multi-level Order Flow Imbalance (OFI) and related microstructure features are used to predict the direction of the next volume bar. Under walk-forward validation, a LightGBM classifier achieves 53.9% mean accuracy and an AUC of 0.547 across five expanding folds, a result statistically distinguishable from a coin flip ($p = 0.023$, one-tailed), while an equivalent logistic regression on the same features is not ($p = 0.155$). We then embed the classifier's probabilities into an inventory-aware Avellaneda–Stoikov market maker. Despite the detectable directional edge, the resulting out-of-sample backtest is unprofitable (net P&L of -31.58 USDT over 705 bars, pseudo-Sharpe -11.67), illustrating that a statistically significant signal need not translate into an economically exploitable one once spread capture, inventory risk, and simplified execution assumptions are taken into account.

1 Introduction

Limit order books record the full state of supply and demand at every price level and update on every order arrival, cancellation, or trade. A long-standing question in market microstructure is whether the order flow visible in the book — as opposed to price history alone — carries information about the direction of the next price move, and whether that information can be turned into an implementable trading strategy. This note is organised around three questions:

1. Does the limit order book contain information predictive of future mid-price movements?
2. Does Order Flow Imbalance (OFI) provide predictive information beyond what a naive linear specification can extract?
3. Can the resulting predictions be used to improve the quoting policy of an inventory-constrained market maker?

The exercise uses only publicly available data from the Binance API and follows a standard research discipline: exploratory characterisation of the data, feature construction grounded in the market microstructure literature, walk-forward (never in-sample) evaluation of predictive models, and an explicit accounting of the assumptions behind the resulting execution strategy. The objective is methodological rigor and honest reporting, not a maximised backtest Sharpe ratio.

2 Data and Market Microstructure

Data were collected via the Binance public REST and WebSocket APIs for the SOL/USDT pair over three sessions (23, 26 and 29 June 2026), each spanning roughly one hour around the NYSE open (14:30 London time), chosen because it is a reliably high-activity window. The local order book is reconstructed to ten levels per side by synchronising REST snapshots with the incremental WebSocket depth-diff stream. In total the sample comprises **173,373** order book update events and **729,293** executed trades.

Two stylised facts motivate the rest of the analysis. First, the best bid-ask spread is essentially constant at the minimum price increment (\$0.01) throughout the session, including through the NYSE open — SOL/USDT behaves as a *large-tick* asset over this window, so spread is not a useful state variable here. A small number of momentarily negative spreads are observed; these are attributable to out-of-order delivery of diff messages on the Binance feed under high message throughput, are not clustered in time, and are excluded without materially affecting the sample. Second, trade count and traded volume both spike sharply at the NYSE open and decay over the following 20–30 minutes, confirming that order flow activity is highly non-uniform in calendar time.

3 Sampling Clocks

Non-uniform activity implies that fixed calendar-time bars contain very different amounts of information depending on when they occur — a form of heteroscedasticity that can obscure genuine structure in a return series. Consistent with this, the autocorrelation function (ACF) of one-second calendar-time returns is statistically indistinguishable from white noise at every lag.

We compare three alternative clocks: a physical clock (10-second bars), an event clock (bars of 300 trades), and a volume clock (bars of 1,500 SOL of cumulative traded volume), each reset at the start of every session. Figure 1 shows the resulting ACFs. Two results stand out. First, excess kurtosis of returns falls sharply moving away from calendar time: **15,968** under the physical clock versus **12.96** (event clock) and **12.55** (volume clock) — most of the extreme fat-tailedness in calendar-time returns is a sampling artefact of mixing quiet and bursty periods, rather than a property of the underlying process. Second, the event and volume clocks show markedly more lags crossing the 95% confidence band than the physical clock, indicating that autocorrelation structure exists but is suppressed by calendar-time sampling. On this basis, all subsequent feature construction and modelling is carried out on the volume clock (1,500 SOL per bar).

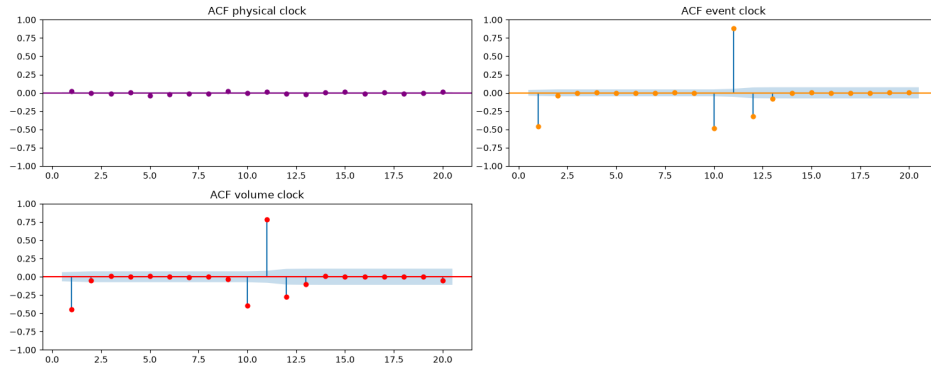


Figure 1: Autocorrelation of returns under the physical (10s), event (300-trade) and volume (1,500 SOL) clocks. The physical clock is indistinguishable from white noise; the event and volume clocks reveal more consistent structure across lags.

4 Feature Engineering

The primary explanatory variable is Order Flow Imbalance (OFI), following Cont, Kukanov and Stoikov (2014). For book level l , define at each update the bid- and ask-side imbalance contributions

$$I_{\text{bid}}^l = \begin{cases} q_{\text{bid}}^l & p_{\text{bid}}^l \uparrow \\ q_{\text{bid}}^l - q_{\text{bid,prev}}^l & p_{\text{bid}}^l \text{ unchanged} \\ 0 & p_{\text{bid}}^l \downarrow \end{cases} \quad I_{\text{ask}}^l = \begin{cases} 0 & p_{\text{ask}}^l \uparrow \\ q_{\text{ask}}^l - q_{\text{ask,prev}}^l & p_{\text{ask}}^l \text{ unchanged} \\ q_{\text{ask}}^l & p_{\text{ask}}^l \downarrow \end{cases}$$

and the multi-level OFI as $\text{OFI} = \sum_{l=1}^{10} (I_{\text{bid}}^l - I_{\text{ask}}^l)$, summed across all events within a volume bar. Level-1-only OFI has mean -1.34 and standard deviation 243.6 across the raw tick sample; the full ten-level version has mean 44.6 and standard deviation $2,539$ — both consistent with a signed, heavy-tailed, approximately mean-zero measure of net book pressure.

Beyond OFI, the feature set includes: queue imbalance at levels 1–5, $qi_l = (q_{\text{bid}}^l - q_{\text{ask}}^l) / (q_{\text{bid}}^l + q_{\text{ask}}^l)$; aggregate depth imbalance over the top five levels; signed trade imbalance from the executed-trade tape; and the two most recent bar-level returns as momentum/reversion controls. All features are winsorised at the 1st/99th percentiles within each fold to limit the influence of outliers without discarding information.

Aggregating to the volume clock yields 983 raw bars; after merging order-book and trade-side features and requiring a complete feature vector, 881 usable bars remain. Removing bars with no net price change (needed to pose next-bar direction as a clean binary problem) leaves **846** bars, split roughly evenly between up-moves (417) and down-moves (429).

A simple Pearson correlation between OFI and the directional target is weak throughout: $r = 0.198$ for the tick-level, level-1 specification, and only $r = 0.059$ once both OFI and the target are moved to the volume clock. This is not strong evidence against a relationship: Pearson correlation captures only linear association, and

thresholding returns into a ± 1 target compresses the dynamic range of the response, mechanically depressing any correlation coefficient even where a real, non-linear relationship exists. This motivates a machine-learning approach.

5 Predictive Modelling and Validation

Because financial time series violate the exchangeability assumption behind random train/test splits, all models are evaluated with *walk-forward* validation: five chronologically ordered, expanding-window folds (TimeSeriesSplit, $k = 5$), so that a model is always tested on data strictly after its training window. An in-sample check — fitting and scoring a univariate logistic regression of next-bar direction on volume-clock OFI on the same data — already produces an unremarkable 53% accuracy and an AUC of 0.538, underscoring that even without out-of-sample discipline this single feature does not come close to a perfect fit; walk-forward evaluation of the same univariate specification gives a mean accuracy of 52.5% (SD 2.2%) and mean AUC of 0.532 (SD 3.0%) — statistically indistinguishable from chance.

Extending the feature set to the full multi-level OFI plus the nine additional microstructure features described in Section 4, two models are compared under the same five-fold walk-forward protocol: a LightGBM gradient-boosted tree ensemble (100 trees, 7 leaves, L1/L2 regularisation) and a logistic regression on standardised features. Table 1 summarises the results, including a one-sample t -test of fold-level accuracy against the 50% no-edge null.

Model	Mean Acc. (SD)	Mean AUC (SD)	t -stat	p (one-tail)	Sig. @ 5%
Logistic Reg. (OFI only)	0.525 (0.022)	0.532 (0.030)	—	—	—
LightGBM (10 features)	0.539 (0.027)	0.547 (0.021)	2.86	0.023	Yes
Logistic Reg. (10 features)	0.540 (0.070)	0.533 (0.082)	1.16	0.155	No

Table 1: Five-fold walk-forward performance on next-volume-bar direction, $n = 846$ bars. Only LightGBM’s fold-level accuracy is statistically distinguishable from 50%.

The point estimates of mean accuracy are close between LightGBM and the enriched logistic regression (53.9% vs. 54.0%), but the logistic specification is roughly **2.5 $\times$ more volatile across folds** (SD 7.0% vs. 2.7%), which is what pushes it below the significance threshold. Two folds (1 and 2) swing from 41% to 60% accuracy for the logistic model, consistent with an unstable linear fit, whereas the LightGBM fold accuracies stay in a narrower 50–58% band. Taken together with the weak linear correlations in Section 4, this pattern is consistent with — though on 846 bars from three sessions of a single asset it cannot be said to *prove* — a directional relationship between multi-level order flow and near-term price direction that is more reliably captured by a non-linear model than a linear one, a finding echoed in the broader OFI literature (Bechler and Ludkovski, 2017; Bugaenko, 2020). With only five folds the test has limited statistical power, and the result should be read as suggestive of a modest, fold-to-fold-fragile edge rather than a strong or robustly stable one.

6 Market Making Strategy: Avellaneda–Stoikov with Signal Integration

The predictive probabilities are integrated into the classical Avellaneda and Stoikov (2008) inventory-aware market-making framework. Given mid-price s , inventory q , risk aversion γ , price volatility σ , arrival intensity k and time remaining $(T - t)$, the reservation price and optimal full spread are

$$r = s - q\gamma\sigma^2(T - t), \quad \delta^* = \underbrace{\gamma\sigma^2(T - t)}_{\text{inventory term}} + \underbrace{\frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{k}\right)}_{\text{adverse-selection term}},$$

with quotes posted at bid $= r - \delta^*/2$ and ask $= r + \delta^*/2$. We further skew the reservation price by the model’s out-of-sample directional probability p_{up} :

$$r_{adj} = r + \alpha(p_{up} - 0.5),$$

so that a bullish signal ($p_{up} > 0.5$) lifts both quotes, capturing more edge on the ask side while reducing exposure to adverse selection on the bid side.

Parameters are calibrated from the data rather than assumed: σ is the realised standard deviation of bar-over-bar mid-price changes on the volume clock ($\sigma = \$0.744$ per bar); k is re-interpreted, since the model operates

in volume time rather than calendar time, as the mean number of trades per volume bar ($k = 820.2$, median 776); $\gamma = 0.1$ and the signal intensity $\alpha = 0.15$ USDT are set exogenously within the ranges typically used in the literature, since risk aversion is a preference parameter rather than an estimable quantity. Figure 2 shows the resulting quotes over the first session with inventory held at zero: the spread starts the session at \$0.0577 (of which \$0.0553 is the inventory/time-decay term and only \$0.0024 is the constant adverse-selection term) and narrows to essentially the selection term alone as the session clock runs down, consistent with the closed-form decomposition of δ^* .

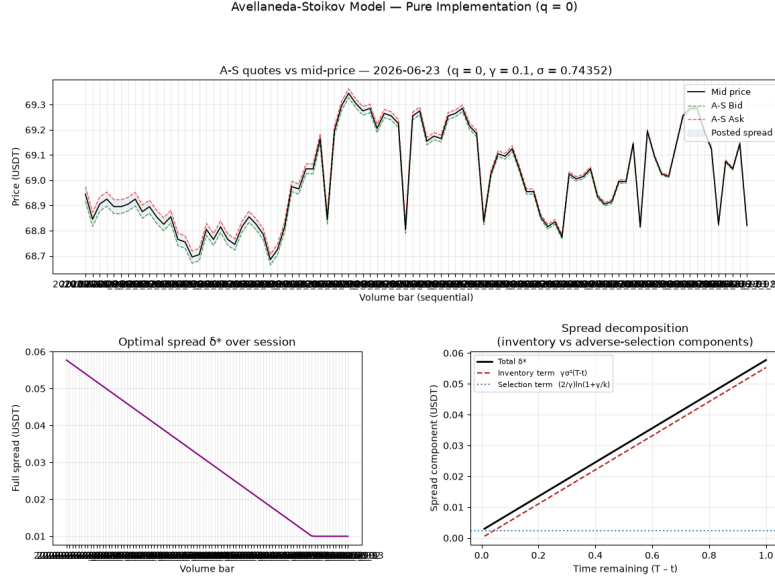


Figure 2: Avellaneda–Stoikov bid/ask quotes around the mid-price over one session (top; $q = 0$, $\gamma = 0.1$), the resulting optimal spread over time (bottom left), and its decomposition into inventory and adverse-selection terms (bottom right).

7 Backtest Design and Results

The strategy is evaluated with an event-driven, bar-by-bar backtest over the 705 volume bars for which a genuine out-of-sample p_{up} is available (i.e., excluding the initial walk-forward training window). Quotes are recomputed at the start of each bar; a fill is assumed only if the *volume-weighted average price of the following bar* crosses the posted quote — a deliberately conservative assumption that ignores queue position and latency. Quote size is fixed at 1 SOL with a hard inventory cap of ± 15 SOL.

Metric	Value
Out-of-sample bars	705
Total fills	536
Net P&L	−31.58 USDT
Spread P&L (execution edge)	−0.19 USDT
Inventory P&L (mark-to-market)	−5.76 USDT
Pseudo-Sharpe (bar-to-bar)	−11.67

Table 2: Out-of-sample backtest performance, Avellaneda–Stoikov market maker with ML signal skew.

Figure 3 decomposes total mark-to-market P&L into its spread and inventory components and plots the resulting inventory path. Despite the statistically detectable directional signal documented in Section 5, the strategy loses money out of sample. We note as a methodological caveat that the reported spread and inventory components do not fully reconcile with the total ($-0.19 - 5.76 \neq -31.58$); the residual reflects the interaction between the price at which a quote is actually filled and the subsequent bar’s mark-to-market price, which the simplified two-way P&L attribution used here does not fully capture and would need to be resolved before this decomposition is used for capital allocation. The headline result, however — that a marginally significant statistical edge did not survive contact with even a stylised, low-latency execution model — is unaffected by that caveat and is the central empirical finding of the exercise.

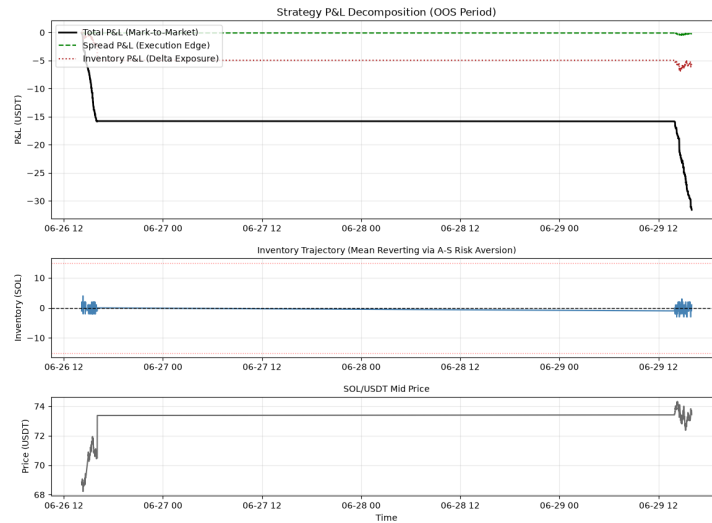


Figure 3: Out-of-sample strategy P&L decomposition (top), inventory trajectory against the ± 15 SOL risk limit (middle), and the underlying mid-price path (bottom).

8 Discussion and Limitations

The results are best summarised as a gap between statistical and economic significance. A non-linear classifier trained on multi-level order flow and related microstructure features produces directional accuracy modestly but measurably above chance (53.9% vs. 50%, $p = 0.023$); an equivalent linear model does not clear the same bar, consistent with a genuinely non-linear, if fragile, relationship. Yet embedding that signal into an inventory-aware market maker under realistic-but-simplified execution assumptions produces a negative out-of-sample P&L. Several limitations bound how far this should be pushed:

- **Sample size and scope.** Three sessions of a single, liquid asset yield only 846 modelling bars and five walk-forward folds; fold-level accuracy is itself noisy (SD 2–7 points), and the significance test has limited power. Generalisation to other assets, liquidity regimes, or time periods is untested.
- **Execution assumptions.** Fills are assumed whenever the following bar’s VWAP crosses a posted quote, with no queue position, latency (real order placement takes on the order of 10–50ms via REST), or partial-fill modelling. In a large-tick market such as SOL/USDT, queue position at the touch is typically the binding constraint on fill probability and is entirely absent here.
- **Volatility estimation.** The realised-volatility estimator used for σ is a simple standard deviation of bar-over-bar price changes and is likely inflated by bid-ask bounce and microstructure noise. A two-scale realised volatility estimator (Zhang, Mykland and Aït-Sahalia, 2005) would be a natural refinement, and would likely allow tighter, more competitive quoting.
- **No market impact or transaction costs.** Quotes are assumed to be filled at the posted price with no slippage and no effect of the market maker’s own flow on subsequent prices; this is a reasonable approximation only at the small (1 SOL) quote size used here.
- **Risk aversion is exogenous.** γ is a preference parameter, not something estimated from data; results are only as good as this (untested) choice, though a sensitivity analysis (not reproduced here for space) shows the qualitative spread/inventory trade-off is stable across $\gamma \in [0.01, 0.5]$.

9 Conclusion

This note set out to test whether limit order book dynamics carry information about short-term price direction and whether that information can be turned into a profitable market-making policy. On three sessions of SOL/USDT data, a volume-clock-sampled, multi-level OFI feature set combined with a gradient-boosted classifier shows a modest but statistically detectable directional edge (53.9% walk-forward accuracy, $p = 0.023$) that a comparable linear model does not reliably reproduce. When that edge is used to skew the quotes of an inventory-aware Avellaneda–Stoikov market maker, however, the resulting out-of-sample backtest is unprofitable under conservative, latency-free execution assumptions. The value of the exercise lies in the reproducible pipeline — data collection, LOB reconstruction, clock selection, feature engineering, leakage-free validation, and an honestly reported execution model — rather than in the backtest return, and the natural next steps are

queue-aware execution modelling, a longer and cross-asset out-of-sample window, and a properly noise-robust volatility estimator, all of which would need to be in place before any of this could be read as evidence of a deployable strategy.

Disclaimer: this note is prepared for educational and research purposes only. Nothing in it constitutes financial advice, investment advice, or a recommendation to trade any instrument.

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